Evolutionary dynamics of eBay matching market

Game Theory
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Abstract

This paper focuses on the evolutionary dynamics of a rich data set describing sequential bargaining occurring in over 25 million listings from eBay's Best Offer platform. We analyze the behavior in these bilateral bargaining situations, where agents interact with very little information about each other's strategies, payoffs, the structure of the games and aspiration levels, and we compare it to the theoretical bargaining literature. The process we deal with is decentralized, as there is no central authority with knowledge about the bargaining agents' parameters. In these settings, the seller randomly encounters other agents (buyers), make bids and offers for potential matches and terminates the game by accepting an offer if the match is profitable, or by rejecting all the offers if no profitable match is found. We analyze the data trying to point out common behaviors and linking existing theories to them. We finally describe a possible type of solution concept as a subgame-perfect Nash equilibrium.

1 Introduction

Matching markets are a common and long-existing form of trade-markets, in which the agents have to choose each other by interacting repeatedly. Many of these markets are decentralized, so the agents do not have any knowledge about the market as a whole. Agents can be sellers and buyers bargaining for the price of a good, but the process is applied in several different everyday situations. It is, for example, the case of workers and firms bargaining with each

other to determine matches and terms of trade [5]. Especially in today's digital age, the theories of the matching market have a big impact on the rising of companies. *Airbnb* is a matching market between travelers and hosts, while *Uber* is a matching market between travelers and drivers. Other examples of matching markets are college admissions and online dating. The American economist *Alvin Roth* has even received a Nobel price in 2012 for using matching markets to match kidneys donors to kidney recipients [15].

Matching markets determine the surplus in all these environments. Analyzing how people match is a question of first-order importance in the real world because the considered agents have aspiration levels that they adjust dynamically based on their experienced payoffs. Over the past years, a large slice of literature in economics has examined several aspects of bargaining in theory and in laboratory experiments. This also includes online markets, which provide the possibility to study real-world setting negotiations with data collected on a giant scale. An example of an online market is given by *eBay*. In this paper, by taking into account a new large dataset regarding *eBay's Best Offer* platform, we analyze the behavior in these bilateral bargaining situations and we try to come up with an equilibrium situation.

The paper is structured as follows. The next section [2] will describe how *eBay's* platform works, to have a better understanding of the bargaining transactions that we later analyze. Section [3] and [4] will introduce some concepts, variables and theories found in the game, while section [5] will analyze the behavioral dynamics of the bargainings. Finally, section [6] will come up with a solution and equilibrium concept.

2 Description of eBay's Best Offer platform

eBay is an American e-commerce corporation for consumer-to-consumer and business-to-consumer transactions. The website allows people to buy and sell a wide variety of goods and services worldwide.

In 2005 the site introduced a new mode of selling through the Best Offer protocol. It

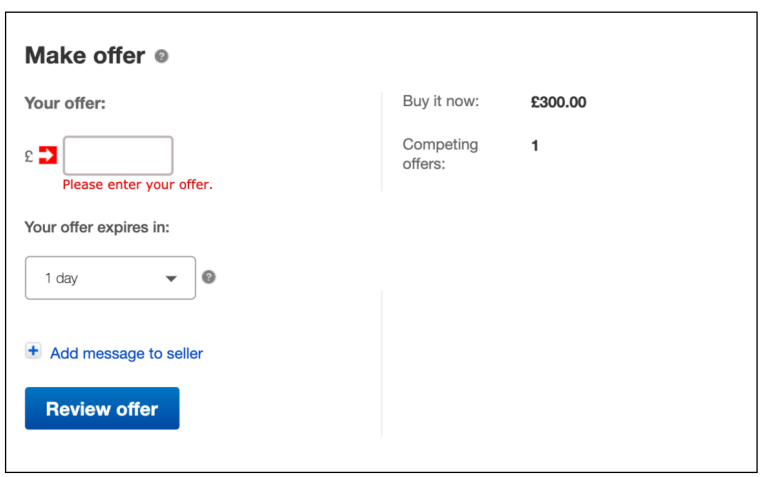


Figure 1: Example of best offer.

consists in an alternating-offer protocol where goods are offered for sale as in a standard fixed-price listing, but also presents an additional option, through the button "Make Offer", that allows the buyer to send an offer to the seller. The last can then accept, reject, or make a counteroffer to the first. In this latter case, the buyer can again accept, decline, or counter. The bargaining keeps going until either party accepts or until the buyer declines. However, if the buyer makes an offer and the seller counters, the buyer can only counter two more times. In case he used all his counteroffers, and he really wants to buy the item immediately, he needs to pay full price, or try again in 30 days if the item is still available. Expired offers, declined offers, and retracted offers count towards a limit of 5 Best Offers per item. Moreover, Best Offers are only good for 48 hours. If the buyer does not receive any feedback from the seller within 48 hours, the offer expires, and it can be considered rejected. In any case, the Best Offer is a commitment to buy, the same as making a bid on an auction item or purchasing an item at full price.

The data set we used was obtained from internal eBay data from Backus at [1]. It consists of over 25 million listings on eBay's Best Offer bargaining platform from June 2012 to December 2013. It includes listing-level variables such as the listing price, reference prices (constructed), category, offers, and counteroffers corresponding to listings.

3 Non-Cooperative Assignment Game

A non-cooperative game is a game with competition between the players and without the players being able to make contracts and group into coalitions. Therefore in a non-cooperative game, the players are competing independently. In our case, a player being a buyer or seller does not gain any personal advantage by making a contract with another player, which is why our scenario falls into this category.

We now introduce some concepts that are needed for the analysis of eBay's matching market. Differently from [2] we analyze matching markets per item. The reason behind this is that the game is played by a seller of a good and by the relative buyers interested in that specific good. Therefore, for our purposes, it doesn't make sense to analyze sellers belonging to different bargaining transactions together.

The population of a game consists of $N = S \cup B$, where $B = \{B_1, B_2, B_3...\}$. The seller S interacts with potential buyers of B that he randomly encounters. Every agent in the game has a **willingness**. Buyers have a **willingness to pay** p^+_{iS}. This expresses the will of B to pay a certain amount of money to be matched with seller S.

$$p^+_{iS} \geq 0 \tag{1}$$

Seller S has a **willingness to accept** q^-_{iS}. This express the will of S to accept a certain amount of money to be matched with buyer B_i.

$$q^-_{iS} \geq 0 \tag{2}$$

As stated earlier, B_i and S are not aware of each others' willingnesses.

The value of a possible match between S and B_i, with $i \in B$ is the potential surplus

$$\alpha_{iS} = (p^+_{iS} - q^-_{iS})_+ \tag{3}$$

For every step k of the game, every agent has an **aspiration level** d^k_i, which expresses a minimal payoff that he is willing to receive in order to be matched. We will discuss of this aspiration level changes depending on the actions of the players in section [6]. For every $i \in B$ we can define the following **assignment function**

$$a_{iS} = \begin{cases} 1, & \text{if } B_i \text{ and S are matched} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

and we can say that B_i and S are matched if $a_{iS} = 1$. Otherwise they are single. When a match is formed, the game ends, and a unique price π_{iS} is taken. This price defines the **payoff** to B_i and S:

$$\phi^k_i = \begin{cases} p^+_{iS} - \pi_{iS}, & \text{if } B_i \text{ and S are matched} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$\phi^k_S = \begin{cases} \pi_{iS} - q^-_S, & \text{if } B_i \text{ and S are matched} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

The match can be **profitable** or not. As [2] defines, a match is profitable if both the buyer and the seller, in expectation, receive a higher payoff when the match is formed.
With this knowledge we can now say that the matching market for a single item is defined by a tuple:

- $\boldsymbol{S}$ is the seller of an item, that leads the bargaining transaction considered.
- $\boldsymbol{B} = \{B_1,\ B_2,\ B_3 ...\}$ is the set of n buyers.
- $\boldsymbol{\alpha} = \big(\alpha_{1S} \dots \alpha_{nS}\big)$ is the vector of matched values.
- $\mathbf{A} = \big(a_{1S}\ ...\ a_{nS}\big)$ is the assignment matrix with 0/1 values.

4 Bargaining theories applicable to eBay

After defining the fundamental variables of a game in the previous section, we can now explore common theoretical models used to reason about sequential bargaining. In this analysis, we consider if agreement or disagreement occurs and when it occurs (immediate or delayed). In doing so we will refer to known bargaining models such as Rubinstein's [4], Perry's [11], and Cramton's [12]. As paper [1] mentions these traditional bargaining models are applicable to our dataset, but are not accurate.

Immediate Agreement
If we consider an immediate agreement scenario where all offers are immediately accepted by the seller, we can reason about it via Rubinstein's [4] model. This consists of a sequence of alternating offers between two players (seller and buyer in our case). Each player is characterized by a cost of bargaining and is omniscient.

This scenario has two main constraints: all interactions should end in agreement and the first offer made is accepted without counteroffers. Of course, this means that under Rubinstein's model the game must end after the first offer is accepted by the opponent. However, as determined empirically by Backus et al. [1] this scenario is consistent only with 32% of observations.

Immediate Disagreement
Shifting our focus now to the disagreement case, we can no longer use Rubinstein's model since it only considers bargaining transactions that converge to a unique solution. This problem can be easily solved by relaxing the assumption under which all players have complete information. This can be done by switching to a model based on incomplete information such as Perry's [11]. Here buyers and sellers have private motivations that are not known.

The equilibrium in this game is reached when one participant makes an offer and the other accepts or rejects but doesn't make a counteroffer. Regarding this case Backus et al. [1] state that this model can rationalize 57% of the sequences.

Delayed Agreement & ***Disagreement***
The last two cases include all the bargaining sequences where agreements or disagreements are reached with a delay. One model that can rationalize both of these instances is Cramton's [12]. Here both parties have private information and motivations similar to Perry's model, however, they also have the conscience to walk away in case no gains can be made from the trade. Theoretically, such a model could explain all the sequences, under the hypothesis that initial offers are not accepted.

This model would be able to explain from 62% to 83% of the cases depending on the assumptions.

5 Behavioral dynamics

In this section, we focus our discussion on some situations that play a prominent role in motivating existing theory. We analyze the formation of a match (agreement) or the rejection of it (disagreement) and how the final negotiated price of the bargaining is reached. The agents we take into consideration are a buyer (B_i), and the seller (S) of a certain good.

1. The seller (S) publishes a product with his list price (Buy-It-Now-Price).

2. The buyer (B_i) is interested in the product:

(a) B_i takes the Buy-It-Now option, transaction ends positive.

(b) B_i makes an offer.

3. If B_i made an offer:

   (a) S can reject.

   (b) S can accept, transaction ends positive.

   (c) S can make a counter offer.

4. If S has rejected:

   (a) The transaction ends negative.

   (b) B_i makes an offer.

   (c) B_i takes the Buy-It-Now option.

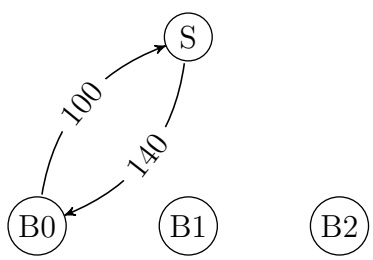
5. If S made an counter offer:

   (a) B_i can accept, transaction ends positive.

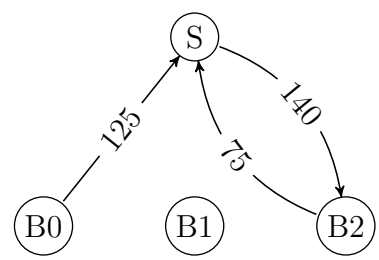
   (b) B_i can make an counter offer.

   (c) B_i rejects, transaction ends negative.

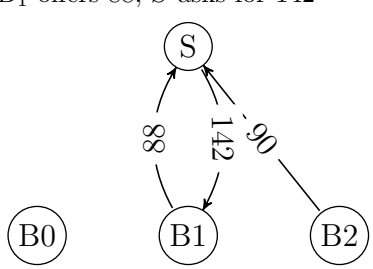
(a) B_0 offers 100, S asks for 140



(b) B_0 offers 125, S refuses.
B_2 offers 75, S asks for 140



(c) B_2 offers 90, S refuses.
B_1 offers 88, S asks for 142



(d) B_1 offers 90, S accepts.

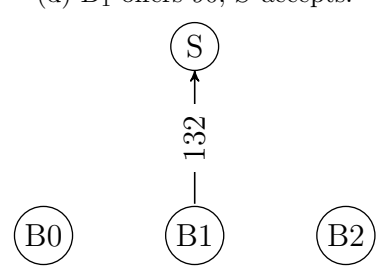


Figure 2: Bargaining sequence example, with one seller S and three buyers.

If the encounter is profitable, meaning that the transaction ends positively, then the pair of S and B matches. Every agent can be matched with multiple agents at the same time. If no new match is formed, S keeps his previous aspiration level (Buy-It-Now price).
An interesting phenomenon observed in Paper [1] that the theories mentioned in section 4 struggle to explain is that of "Split the Difference" kind of offers. There, it is studied how prices evolve in a bargaining fight of type offer-counter offer, which we write

$$(p_k)_{k=1}^{7} \text{ where } p_k = \text{offer at time } k$$

at step $k = 0$ the seller chooses a "Buy It Now" price p_0, at step $k = 1$ the buyer then places and offer p_1 and so forth for the subsequent steps.
As demonstrated by [1] players tend to make more favorable offers for the opposing player every round. In fact, it is very rare that a seller remains at his Buy-It-Now price for several rounds, 98% of them concede to the buyer already with the first counteroffer. Following Table VII in paper [1] we can analyze the amount of concession every round. When dealing with such a game, one can define a useful parameter γ_k, uniquely determined by the relation

$$p_k = \gamma_k p_{k-1} + (1 - \gamma_k) p_{k-2}$$

which is to be regarded as the weight the player places on her own previous offer, for example

$$p_1 = \gamma_1 p_0 + (1 - \gamma_1) 0 \rightsquigarrow \gamma_1 = p_1/p_0$$

in this case, γ_1 could be interpreted as her bliss point of paying nothing for the traded good. This index in some sense encodes how the change in aspiration evolves at each stage of the game and provides an empirical motivation for choosing a simple, one-directional learning model as we will do in Section 6. Therefore we use γ_{k-1} as an indicator of how much the opposing player conceded in the previous round and γ_k to illustrate a player's own approach to the counterpart.

When γ_{k-1} is positive, a player approached the other, where higher numbers mean greater approach. Table VII also suggests that it is irrelevant whether a product is new or used. Backus et al. [1] come to the conclusion that the highest concession is usually made by the seller in period 4 whereas buyers react with their greatest concession in period 5. This pattern, that higher concession from one player in round $k-1$ results in a higher concession in round k from the opposing player, can be seen throughout the data.

The analysis of the weights γ_k in games of type offer-counteroffer ending at time k has been performed via histograms, see Figure 9 in Paper [1], and reveals that:

1. The first offer tends to be close to the BIN (Buy it now) price, equivalently γ_1 takes mostly high values ($\gamma_1 \sim 0.7$).

2. The "Split the Difference" or mid-point offer ($\gamma_t \sim 0.5$) tends to be the modal offer at least in the early-middle stages of the game and is a common mass point at different steps.

3. In the late stage of the game the modal gives almost zero weight to the most recent offer of the opponent, which can be interpreted as the fact that the players are willing to modify their previous offers less and less. Immediately after this choice still comes the already mid-point one, as the most common.

The phenomenon of splitting the *surplus* is well known in the literature and is usually attributed to some notion of fairness associated to players behavior, see [7] for a theoretical study in which is not fairness by itself but the desire of a player to be perceived as fair from the others which are analyzed. Here, however, the information available to players might be incomplete and thus there is no a priori notion of surplus to be considered, what the data suggests is that the mere fact of "choosing the middle" seems more appealing regardless of any possible future money return.

5.1 Bargaining Power

Following [1] we want to give a measure for the common belief that more experienced players participating in a bargaining game should end up with a bigger share of the generated surplus. In abstract bargaining models this "expertise" can either appear as a primitive e.g. Rubinstein in [4] defines a "discount factor" describing how much a player is willing to adapt his offer at each turn of the bargaining game, in some sense measuring its patience, or be deduced as a reduced-form feature dependent from the given primitives.

Taking up the patience point of view on bargaining power, authors of [1] mark those who choose the slowest shipping method after winning the bargain as those who may have more bargaining power. Also, a measure of cumulated expertise can be given by looking at the number of won games, and be determined directly from the data on Table 1 in Paper [1].

Consistently with Table 2 in Paper [1] the statistical relevance of both patience and experience (buyer fixed effects) highly depends on whether the bargained product is new or used. There, it is shown that buyer fixed effects influence over 83% the variance in prices while only 45-60% is determined by sellers fixed effects and most importantly the product identity is substantially more relevant for predicting prices in new products. This means that statistically, the "ability" of a player counts more for buyers of used goods rather than sellers in general or buyers of new products. This reflects the fact that new products tend to have a better defined reference price. Performing a regression analysis of the data, where the dependent variable is the normalized price and the regressors are buyer and seller attributes, authors of [1] were able to obtain Table 3. This strengthens the observations coming from Table 2, in particular in Column 5 it can be seen that 1% point of change in buyers experiences results in 1.8% decrease with respect to the reference price. When also considering patience as did in Column 6, the influence on reference price increases to 2.8%. For sellers, the experience is even more effective leading to an increase up to 6.4%, however when also controlling for patience this increase vanishes, see column 6 again.

Overall the conclusion is that experienced agents obtain higher yields, which seems a trivial observation, however, the point of [1] is exactly to support this "common knowledge" with data. Experienced buyers tend to save around 0.5% in the early stages of the bargaining

and over 2% from three to four bargaining steps, whereas experienced sellers obtain the highest surplus, namely up to 2%, during the first two offers, and yield 0.2% more in the following stages.

6 Solution Concept

This section is devoted to our analysis of the data found in [1]. Motivated by what explained in Section 4 and subsequent we base our study on simple directional learning. We first briefly describe the setup, then present our findings with a series of plots. The first set of four describes the evolution of offers and request, the last two give a global overview of the game outcomes.

There are some considerations to be mentioned before analyzing these data. As the aspiration adjustment theory [3] mentions, we assume a simple directional learning model: a single agent S tends to lower his offer in the hope of matching with buyers, while a single agent B_i tends to increase his bids. This approach is in contrast with traditional theories described in Section 4 due to its cooperative nature, nonetheless it lets us understand this common behavior found in online markets.

We will describe the process in detail. We assume the active agent to be a buyer B_i drawn at random with $i \in B$, therefore unique at every step. In any step k, we assume that the two players' bids are chosen such that the resulting payoff to each player is at least equal to his aspiration level, and with positive probability is exactly equal to his aspiration level. A bid is rejected if it is not profitable for both players.

One could (following ideas from [2]) set this up formally as follows. Assume buyer B_i meets seller S at time k. Let p_i^k be the offer of B_i and q_i^k the minimal amount of money the seller is willing to accept from player i for the (fixed) bargained good. We can describe those two variables at time k as made of a random and a deterministic part. The deterministic part is given by the aspiration level at the previous time d_i^{k-1} while the random part represents an "exogenous shock" with this we mean, the random event that a player achieves more than desired (pay less, or sell for more money), this is modeled via a pair of random independent variables P_i^k, Q_i^k taking values in $\delta \cdot \mathbb{N}_0$. The bids at step k, are then given by:

$$p_i^k = (p_{iS}^+ - d_i^{k-1}) - P_i^k \qquad \text{and} \qquad q_i^k = (q_{iS}^- + d_i^{k-1}) + Q_i^k$$

We define a game as ended if the object has been sold. Given the nature of our case, we see clearly that after a player makes a bid there are only two possible outcomes: the bid is accepted by the other player and the game ends, or the bid is not accepted and the game continues. If a bid is rejected we assume that with positive probability the two players reduce their aspiration level. Note that this is consistent with experimental data see the discussion above. Formally, if we take an independent random variable X^{k+1} that takes values in $\delta \cdot \mathbb{N}_0$, where δ is some minimal unit of currency, we can express the change in the aspiration level of B_i and S like this:

$$d_i^{k+1} = (d_i^k - X_i^{k+1})_+ \qquad \text{and} \qquad d_S^{k+1} = (d_S^k - X_S^{k+1})_+$$

This behavior is clearly visible in figure 3, where for each step we take the difference between the new offer and the previous one for the buyer, and the new request price and the previous one for the seller. This is directly linked to changes in their aspiration levels. We considered 10000 bargaining interactions between an individual seller and a buyer and then took the average of the change in offer. The bargaining sequences we plot are made of a maximum of 5 and 6 total turns.

In figure 4 we also show behaviors sampled from the data set, representing accurate percentages of actions in the bargaining sequences. The left image represents bargaining transactions ended without selling the item, the right one instead illustrates sold items situations. The first thing that stands out from the right image is that most of the time, when an item is sold, both of the agents involved (seller S and buyer B_i) are open to change their bids in order to reach a match. Instead, when an item is not sold, it is more common that the buyer increases his offer (33.7 %) than the seller decreasing his request price; indeed the percentage of both changing is lower than before, going from 72.8% to 55.4%. By considering that the total number of sold items in our transactions is 10277 and that the total number of unsold items is 19786, we can say that it's more frequent than only the buyer increases his offers.

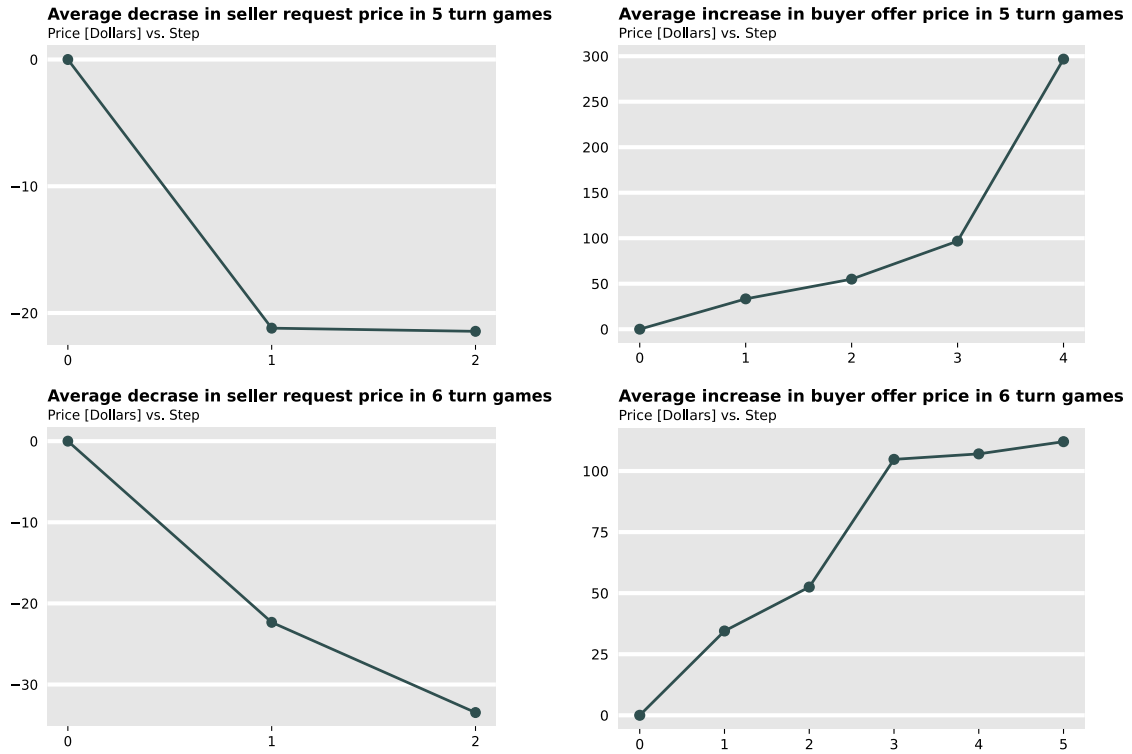


Figure 3: Average change in offers considering 10000 transactions of length 5 and 6.

The code used to generate the plots can be found on: https://github.com/zommi/ebay_matching_market_analysis

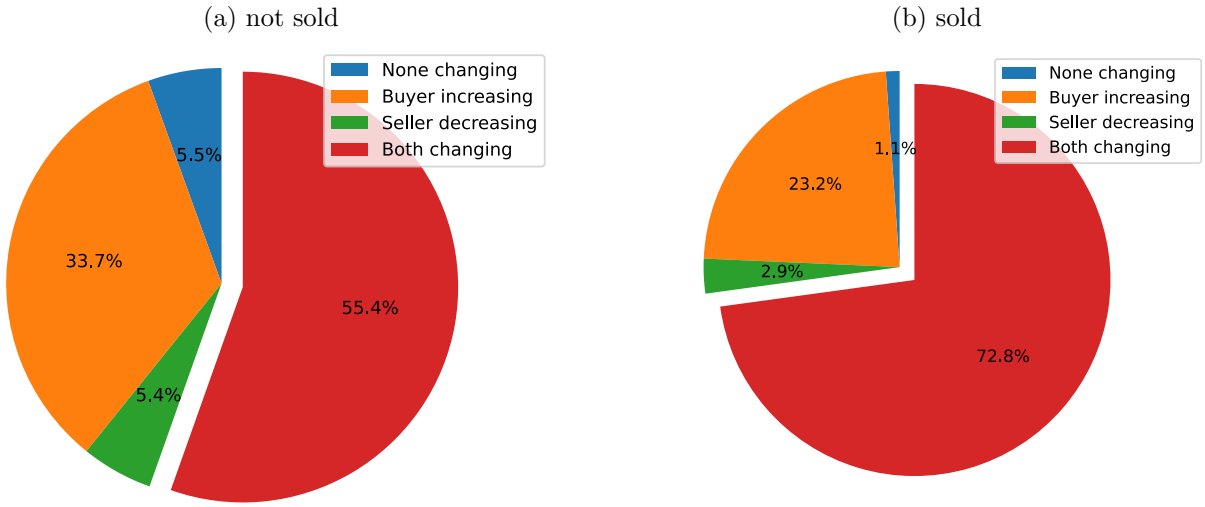


Figure 4: Percentages of behaviors sampled from 30035 bargaining sequences of N turns with $3 \leq N < 15$.

6.1 Nash Equilibrium

In this paper we focused on the case of one seller and several buyers. The target of this non-cooperative theory of bargaining is to find theoretical predictions of what agreement will be reached by the bargainers. With this, it is possible to explain how the bargaining outcome depends on the parameters of the bargaining problem.
At every step in time the seller is interacting with only one buyer. Suppose the game between a buyer B_i and the seller S can be summarized, at any step k, like this:

B_i\S	Sells	Does not sell
Buys	$[\phi^k_{iS},\ \phi^k_S]$	$[\star,\ 0]$
Does not buy	$[0,\ \phi^k_S]$	[0,0]

Table 1: Bymatrix game played by buyer B_i and seller S at time k.

The payoffs are shown in the table. When the buyer B_i buys the item from a different seller than S, we indicate his payoff with $\star$ because we do not consider that situation.
We want to define the kind of equilibrium concept we will adopt in the analysis of our game. In particular, we embrace the so called "Nash Program" and implement a sort of cooperative solution concept in our non-cooperative setting. In the standard literature this approach is known as **Nash Bargaining Solution** and first appeared in Paper [13].
Before introducing it, we need to set up some notation and give some definitions, in doing this we closely follow [17].
One calls a convex, compact subset $\mathcal{S}$ of $\mathbb{R}^2$ a **bargaining set** if in addition it has a **thread**

point, that is, a point $(u_0, v_0) \in \mathcal{S}$ such that

$$(u, v) \in \mathcal{S} \Longrightarrow u \geq u_0,\ v \geq v_0$$

Given a bargaining set $\mathcal{S}$ one stipulates the following axioms for the Nash Bargaining Equilibrium:

1. **(Admissibility)** $N(S) = (U, V)^T \in \mathcal{S}$
2. **(Pareto Optimality)**

$$(u, v)^T \in \mathcal{S} \text{ with } u \geq U,\ v \geq V \Longrightarrow (u, v) = (U, V)$$

3. **(Invariance of Utility Scaling)** for any $a, b, c, d \in \mathbb{R}$ with $a, c > 0$

$$\mathcal{S}' = \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \cdot S + \begin{pmatrix} b \\ d \end{pmatrix} \Longrightarrow N(\mathcal{S}') = \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \cdot \begin{pmatrix} U \\ V \end{pmatrix} + \begin{pmatrix} b \\ d \end{pmatrix} = (a \cdot U + b, c \cdot V + d)$$

4. **(Symmetry)**

$$\mathcal{S} \text{ symmetric} \Longrightarrow N(\mathcal{S}) = (U, V) = (U, U)$$

5. **(Irrelevance of Unused Alternatives)** If $\mathcal{S}, \mathcal{T}$ are bargaining sets with the same thread point and $\mathcal{S} \subset \mathcal{T}$ then

$$N(\mathcal{T}) \notin \mathcal{S} \text{ or } N(\mathcal{T}) = N(\mathcal{S})$$

Under the above axioms a unique Nash equilibrium $N(\mathcal{S})$ can be shown to exist and can be explicitly computed by maximization of the **Nash Product**

$$(U - u_0) \cdot (V - v_0)$$

with (u_0, v_0) the thread point of $\mathcal{S}$. For a proof of this fact see for example [17].
With these axioms at hand, and the existence of such an equilibrium, we can now reason about our scenario. We face two main difficulties. The above type of equilibrium works well with two players whereas we have several. Moreover, our setting is dynamic, it is played in several rounds: we are not just "splitting a pie" between two people but rather every person can make an offer for a piece of this pie in a given turn and the owner can only choose one. This person, which corresponds to the seller, is selfish and over time learns by calling one of them at random which one is "best" (according to her current aspiration level) to share it with. Taking $\mathcal{S}$ to be the set given by the bi-matrix game contained in Table 1 we construct

$$\mathcal{S}_i^k = \{(\phi_{iS}^k(x), \phi_S^k(y)) \in \mathbb{R}^2 : 0 \leq x, y \leq \text{BIN price }\}$$

Note that our payoffs are linear in the price e.g.

$$\phi_{iS}(x) = p_{iS}^+ - x$$

and thus concave so that the theory applies. Those sets are random since they depend on the specific buyer activated at time k.

By the classical theory briefly described above this gives a random Nash bargaining equilibrium at each time step k (that is a distribution of Nash Bargaining equilibria over the set of buyers). To solve the fact that the game is dynamic, we proceed by **backward induction**: since we have a finite number of players and thus a finite maximum number of offers/counteroffers the game is finite and we can determine a random subgame-perfect Nash equilibrium as follows. Assuming a sequence of bargaining players is given, one can then construct a game tree representing the possible outcomes in the game. At each node of the tree, it is possible to determine the bargaining Nash equilibrium as described above. The subgame-perfect Nash equilibrium is then found by choosing the most favorable for the seller going backward in the tree. Note that, by definition of the game each sequence of players is chosen randomly. Hence the tree itself is random and we get a random subgame-perfect Nash equilibrium for our game.

7 Conclusions

In this paper, we analyzed behaviors coming from bargaining sequences of a massive data set coming from *eBay*, used by millions of people all over the world. We organized the kind of transaction we found on the data in different sets based on similar patterns. From these sets, we reported accurate percentages of price increases and decreases from both sellers and buyers sides. We explained how the aspiration level of the agents involved in the bargaining changes at any step, depending on the bids and offers made in previous steps, and we showed data where this process can be seen. By taking into account different existing theories, we showed which ones didn't work and which ones could be found and adapted in our settings. We finally described a possible type of solution concept as a subgame-perfect Nash equilibrium. An interesting possible development in these directions could be learning the distributions of the offers along with the random game trees, defined in Section 6.1 from the data.

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